Zaira Ramos de la Cruz
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Ingeniería en Software Virtual

Introducción

En la actualidad, los servicios web son fundamentales para el funcionamiento de aplicaciones y sistemas en red, ya que permiten la comunicación entre clientes y servidores a través del protocolo HTTP. La correcta instalación y configuración de estos servicios es una habilidad esencial en el área de sistemas y redes, especialmente en entornos donde se requiere disponibilidad, escalabilidad y compatibilidad entre diferentes plataformas.

En esta práctica se llevó a cabo la instalación de diversos servicios HTTP tanto en sistemas operativos Windows como Linux, incluyendo servidores ampliamente utilizados como Apache, Nginx, Tomcat e IIS. Además, se desarrolló un script automatizado en cada entorno con el propósito de simplificar el proceso de instalación y configuración, asegurando que los servicios pudieran ejecutarse simultáneamente mediante la asignación de puertos específicos. Esto permitió no solo comprender el funcionamiento individual de cada servicio, sino también su convivencia dentro de un mismo sistema.

Objetivo

Instalar y configurar diferentes servicios HTTP en los sistemas operativos Windows y Linux, mediante el uso de scripts automatizados que permitan su correcta implementación y ejecución simultánea, asignando puertos específicos a cada servicio, con el fin de comprender su funcionamiento, administración y compatibilidad en distintos entornos.

Datos del entorno

Para la realización de la práctica se utilizaron máquinas virtuales desplegadas en VirtualBox, configuradas dentro de una red interna que permitió la comunicación directa entre los nodos sin dependencia de una red externa.

Los sistemas operativos utilizados fueron:

- Servidor Linux: Ubuntu Server (sin entorno gráfico, administración por CLI)

- Servidor Windows: Windows Server
- Cliente: Ubuntu Desktop para pruebas de conexión

Bitácora de desarollo

Linux

Se configuraron tres servicios HTTP en un servidor Linux: Apache, Tomcat y Nginx, asignando un puerto distinto a cada uno para permitir su ejecución simultánea.
Se verificó su funcionamiento mediante pruebas de acceso desde el navegador y la validación de puertos activos, confirmando que cada servicio respondía correctamente en su respectivo puerto.

Apache

- **Instalación**

  sudo apt update

```
zai@srv-linux-sistemas:~$ sudo apt update
Obj:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Obj:2 http://archive.ubuntu.com/ubuntu noble InRelease
Obj:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Obj:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias... Hecho
Leyendo la información de estado... Hecho
Se pueden actualizar 30 paquetes. Ejecute «apt list --upgradable» para verlos.
zai@srv-linux-sistemas:~$ _
```

  sudo apt install apache2 -y

```
ervice restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

o containers need to be restarted.

ser sessions running outdated binaries:
zai @ session #1: login[902]
```

  Iniciar apache

```
zai@srv-linux-sistemas:~$ sudo systemctl start apache2
```

Verificar estado

systemctl status apache2

```
zai@srv-linux-sistemas:~$ systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Fri 2026-04-24 22:47:36 UTC; 6s ago
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 21029 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 21045 (apache2)
      Tasks: 55 (limit: 1055)
     Memory: 5.2M (peak: 5.3M)
        CPU: 35ms
     CGroup: /system.slice/apache2.service
             ├─21045 /usr/sbin/apache2 -k start
             ├─21047 /usr/sbin/apache2 -k start
             └─21048 /usr/sbin/apache2 -k start
```

- **Configuración de puerto (8081)**

```
# If you just change the port or add more ports here, you will likely also
# have to change the VirtualHost statement in
# /etc/apache2/sites-enabled/000-default.conf

Listen 8081

<IfModule ssl_module>
        Listen 443
</IfModule>

<IfModule mod_gnutls.c>
        Listen 443
</IfModule>
```

VirtualHost modificado

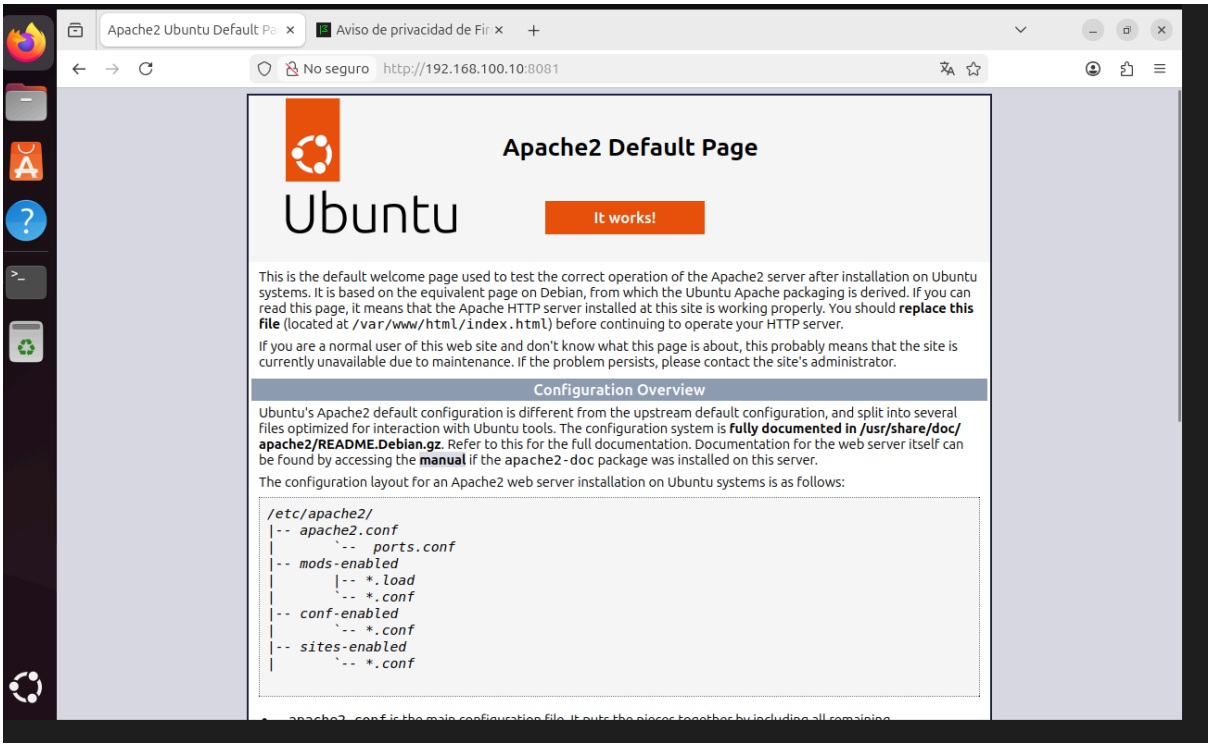
```
<VirtualHost *:8081>
        # The ServerName directive sets the request scheme, hostname and port that
        # the server uses to identify itself. This is used when creating
        # redirection URLs. In the context of virtual hosts, the ServerName
        # specifies what hostname must appear in the request's Host: header to
        # match this virtual host. For the default virtual host (this file) this
        # value is not decisive as it is used as a last resort host regardless.
        # However, you must set it for any further virtual host explicitly.
        #ServerName www.example.com

        ServerAdmin webmaster@localhost
        DocumentRoot /var/www/html

        # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
        # error, crit, alert, emerg.
        # It is also possible to configure the loglevel for particular
        # modules, e.g.
        #LogLevel info ssl:warn

        ErrorLog ${APACHE_LOG_DIR}/error.log
        CustomLog ${APACHE_LOG_DIR}/access.log combined

        # For most configuration files from conf-available/, which are
        # enabled or disabled at a global level, it is possible to
        # include a line for only one particular virtual host. For example the
        # following line enables the CGI configuration for this host only
        # after it has been globally disabled with "a2disconf".
        #Include conf-available/serve-cgi-bin.conf
</VirtualHost>
```

Validación de configuración

```
zai@srv-linux-sistemas:~$ sudo ss -tuln | grep 8081
tcp   LISTEN 0      511                                    *:8081            *:*
```

- Pruebas

Prueba desde cliente

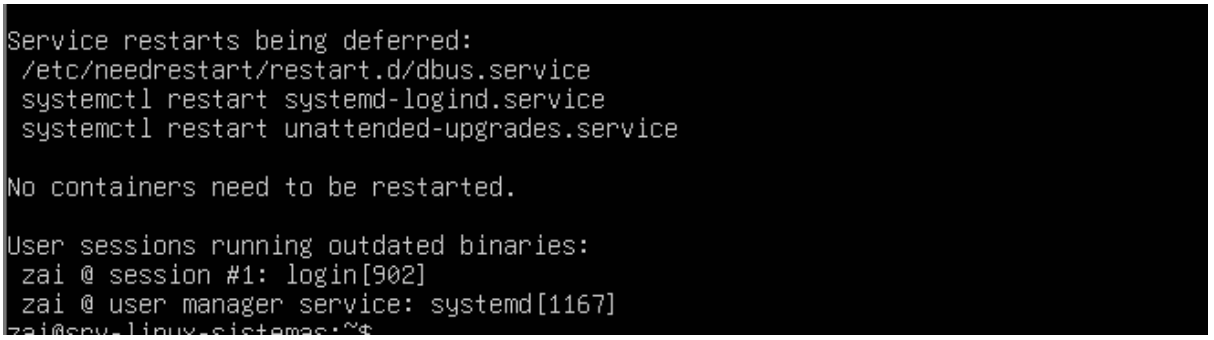


Tomcat

- **Instalación**

  sudo apt update

  sudo apt install tomcat10 -y



- **Configuración del puerto (8082)**

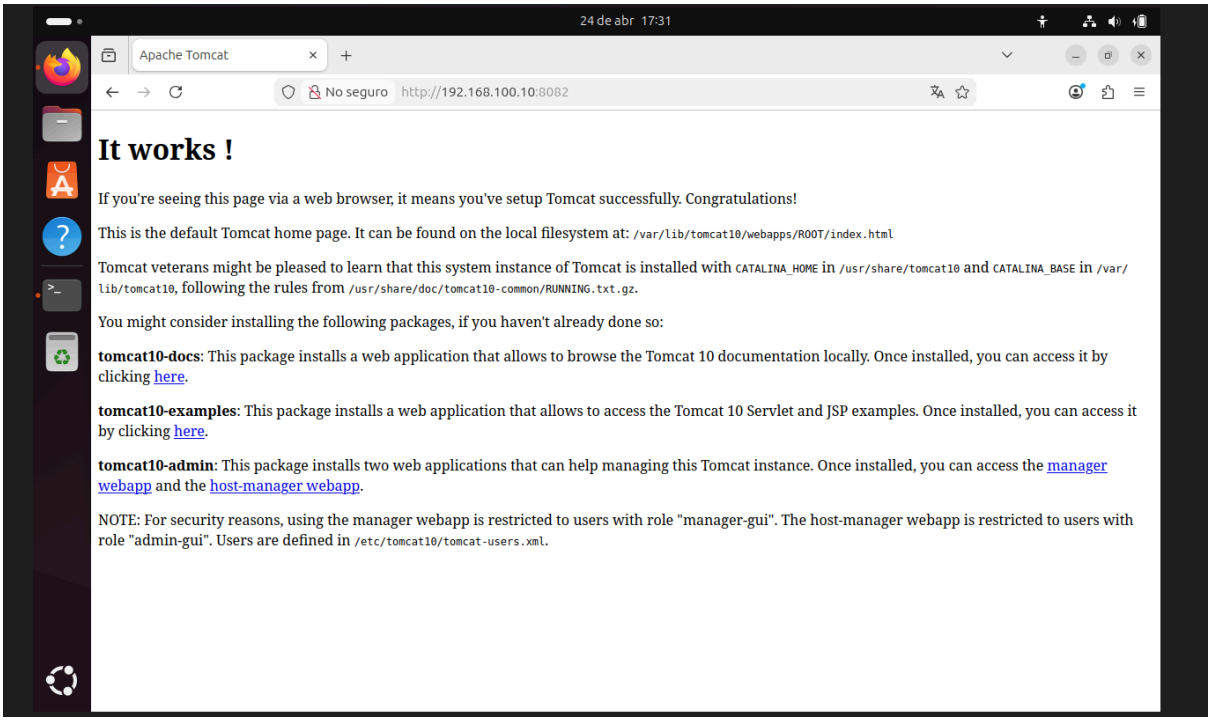
```
-->
<Connector port="8082" protocol="HTTP/1.1"
           connectionTimeout="20000"
           redirectPort="8443"
           maxParameterCount="1000"
           />
<!-- A "Connector" using the shared thread pool-->
<!--
<Connector executor="tomcatThreadPool"
           port="8082" protocol="HTTP/1.1"
           connectionTimeout="20000"
           redirectPort="8443"
           maxParameterCount="1000"
           />
-->
```

Verificación

```
zai@srv-linux-sistemas:~$ sudo systemctl restart tomcat10
zai@srv-linux-sistemas:~$ sudo ss -tuln | grep 8082
tcp   LISTEN 0      100                                  *:8082            *:*
```

- **Prueba**

Prueba desde cliente



Nginx

- **Instalación**

  sudo apt install nginx -y

```
Service restarts being deferred:
 /etc/needrestart/restart.d/dbus.service
 systemctl restart systemd-logind.service
 systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
 zai @ session #1: login[902]
 zai @ user manager service: systemd[1167]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
```

- Configuración de puerto (8083)

  sudo nano /etc/nginx/sites-available/default

```
##
# You should look at the following URL's in order to grasp a solid understanding
# of Nginx configuration files in order to fully unleash the power of Nginx.
# https://www.nginx.com/resources/wiki/start/
# https://www.nginx.com/resources/wiki/start/topics/tutorials/config_pitfalls/
# https://wiki.debian.org/Nginx/DirectoryStructure
#
# In most cases, administrators will remove this file from sites-enabled/ and
# leave it as reference inside of sites-available where it will continue to be
# updated by the nginx packaging team.
#
# This file will automatically load configuration files provided by other
# applications, such as Drupal or Wordpress. These applications will be made
# available underneath a path with that package name, such as /drupal8.
#
# Please see /usr/share/doc/nginx-doc/examples/ for more detailed examples.
##

# Default server configuration
#
server {
        listen 8083 default_server;
        listen [::]:8083 default_server;

        # SSL configuration
        #
        # listen 443 ssl default_server;
        # listen [::]:443 ssl default_server;
        #
        # Note: You should disable gzip for SSL traffic.
        # See: https://bugs.debian.org/773332
        #
        # Read up on ssl_ciphers to ensure a secure configuration.
        # See: https://bugs.debian.org/765782
        #
        # Self signed certs generated by the ssl-cert package
        # Don't use them in a production server!
        #
        # include snippets/snakeoil.conf;

        root /var/www/html;

        # Add index.php to the list if you are using PHP
        index index.html index.htm index.nginx-debian.html;
```

verificación de puerto

sudo ss -tuln | grep 8083

```
zai@srv-linux-sistemas:~$ sudo ss -tuln | grep 8083
tcp   LISTEN 0      511                                  0.0.0.0:8083      0.0.0.0:*
tcp   LISTEN 0      511                                     [::]:8083         [::]:*
```

- **Pruebas**

Prueba desde cliente

Ejecución script Automatizado

```
ai@srv-linux-sistemas:/media/sf_Practica_HTTP/scripts$
ai@srv-linux-sistemas:/media/sf_Practica_HTTP/scripts$ chmod +x linux_http.sh
ai@srv-linux-sistemas:/media/sf_Practica_HTTP/scripts$ ./linux_http.sh
INFO] Verificando Apache...
INFO] Apache ya instalado
H00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'Se
s message
yntax OK
OK] Apache activo en puerto 8081
INFO] Verificando Tomcat...
INFO] Tomcat ya instalado
OK] Tomcat activo en puerto 8082
INFO] Verificando Nginx...
INFO] Nginx ya instalado
OK] Nginx activo en puerto 8083
INFO] Validando puertos...
OK] Puerto 8081 activo
WARN] Puerto 8082 no activo
OK] Puerto 8083 activo
INFO] Proceso finalizado
ai@srv-linux-sistemas:/media/sf_Practica_HTTP/scripts$ _
```

Windows

Se configuraron tres servicios HTTP en un servidor Windows: IIS, Nginx y Apache asignando un puerto distinto a cada uno para permitir su ejecución simultánea.

Se verificó su funcionamiento mediante pruebas de acceso desde el navegador y la validación de puertos activos, confirmando que cada servicio respondía correctamente en su respectivo puerto.

IIS

Verificación de IP

ipconfig

```
PS C:\Users\Administrator> ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : fd17:625c:f037:2:97cc:5ad4:bdc5:3c54
   Link-local IPv6 Address . . . . . : fe80::1214:ba50:7ad3:26d8%8
   IPv4 Address. . . . . . . . . . . : 10.0.2.15
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::2%8
                                       10.0.2.2

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::af47:25c1:8e98:b575%15
   IPv4 Address. . . . . . . . . . . : 192.168.100.20
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
PS C:\Users\Administrator> 192.168.100.20
PS C:\Users\Administrator>
```

Prueba de conectividad de Windows a cliente

ping 192.168.100.30

```
PS C:\Users\Administrator> ping 192.168.100.30

Pinging 192.168.100.30 with 32 bytes of data:
Reply from 192.168.100.30: bytes=32 time<1ms TTL=64
Reply from 192.168.100.30: bytes=32 time<1ms TTL=64
Reply from 192.168.100.30: bytes=32 time=1ms TTL=64
Reply from 192.168.100.30: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.100.30:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
PS C:\Users\Administrator>
```

- **Instalación**

  Install-WindowsFeature -Name Web-Server -IncludeManagementTools}

```
PS C:\Users\Administrator> Install-WindowsFeature -Name Web-Server -IncludeManagementTools

Success Restart Needed Exit Code      Feature Result
------- -------------- ---------      --------------
True    No             NoChangeNeeded {}
```

  Verificación

  Get-Service W3SVC

```
PS C:\Users\Administrator> Install-WindowsFeature -Name Web-Server -IncludeManagementTools

Success Restart Needed Exit Code      Feature Result
------- -------------- ---------      --------------
True    No             NoChangeNeeded {}
```

- **Configuración del puerto (8084)**

  Crear sitio en el puerto 8084

  Crear carpeta

  mkdir C:\sitio_iis

```
PS C:\Users\Administrator> mkdir C:\sitio_iis


    Directory: C:\


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        4/24/2026   7:52 PM                sitio_iis
```

  Crear página

  echo "<h1>IIS funcionando en puerto 8084</h1>" > C:\sitio_iis\index.html

```
PS C:\Users\Administrator> echo "<h1>IIS funcionando en puerto 8084</h1>" > C:\sitio_iis\index.html
```

  Crear sitio IIS

  Import-Module WebAdministration

  New-Website -Name "IIS_8084" -Port 8084 -PhysicalPath "C:\sitio_iis"

```
PS C:\Users\Administrator> Import-Module WebAdministration
PS C:\Users\Administrator> Get-Command New-Website

CommandType     Name                                               Version    Source
-----------     ----                                               -------    ------
Cmdlet          New-Website                                        1.0.0.0    WebAdministration


PS C:\Users\Administrator> New-Website -Name "IIS_8084" -Port 8084 -PhysicalPath "C:\sitio_iis"

Name             ID   State      Physical Path                 Bindings
----             --   -----      -------------                 --------
IIS_8084         3    Started    C:\sitio_iis                  http *:8084:
```

Abrir firewall

New-NetFirewallRule -DisplayName "IIS 8084" -Direction Inbound -Protocol TCP -LocalPort 8084 -Action Allow

```
S C:\Users\Administrator> New-NetFirewallRule -DisplayName "IIS 8084" -Direction Inbound -Protocol TCP -LocalPort 8084 -Action Allow


ame                   : {74996195-0401-47c5-9992-485dbe58cae8}
isplayName            : IIS 8084
escription            :
isplayGroup           :
roup                  :
nabled                : True
rofile                : Any
latform               : {}
irection              : Inbound
ction                 : Allow
dgeTraversalPolicy    : Block
ooseSourceMapping     : False
ocalOnlyMapping       : False
wner                  :
rimaryStatus          : OK
tatus                 : The rule was parsed successfully from the store. (65536)
nforcementStatus      : NotApplicable
olicyStoreSource      : PersistentStore
olicyStoreSourceType  : Local
```

- **Prueba**

  Desde cliente

Apache

- Instalación

```
PS Microsoft.PowerShell.Core\FileSystem::\\VBOXSVR\Practica_HTTP> copy httpd-2.4.66-260223-Win64-VS18.zip C:\
PS Microsoft.PowerShell.Core\FileSystem::\\VBOXSVR\Practica_HTTP> cd C:\
PS C:\> ls


    Directory: C:\


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----          4/8/2026   2:45 PM                FTP
d-----         3/29/2026  10:49 PM                inetpub
d-----         4/24/2026   9:04 PM                nginx
d-----         11/5/2022  12:03 PM                PerfLogs
d-r---         3/17/2026  12:36 AM                Program Files
d-----         9/15/2018   2:08 AM                Program Files (x86)
d-----         4/24/2026   7:54 PM                sitio_iis
d-r---          3/3/2026   5:15 PM                Users
d-----         3/29/2026  10:49 PM                Windows
-a----         4/24/2026  10:00 PM       12390388 httpd-2.4.66-260223-Win64-VS18.zip


PS C:\> Expand-Archive httpd-2.4.66-260223-Win64-VS18.zip
PS C:\> ls


    Directory: C:\


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----          4/8/2026   2:45 PM                FTP
d-----         4/24/2026  10:40 PM                httpd-2.4.66-260223-Win64-VS18
d-----         3/29/2026  10:49 PM                inetpub
d-----         4/24/2026   9:04 PM                nginx
d-----         11/5/2022  12:03 PM                PerfLogs
d-r---         3/17/2026  12:36 AM                Program Files
d-----         9/15/2018   2:08 AM                Program Files (x86)
d-----         4/24/2026   7:54 PM                sitio_iis
d-r---          3/3/2026   5:15 PM                Users
d-----         3/29/2026  10:49 PM                Windows
-a----         4/24/2026  10:00 PM       12390388 httpd-2.4.66-260223-Win64-VS18.zip


PS C:\> cd httpd-2.4.66-260223-Win64-VS18
PS C:\httpd-2.4.66-260223-Win64-VS18> ls


    Directory: C:\httpd-2.4.66-260223-Win64-VS18


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         4/24/2026  10:41 PM                Apache24
-a----         1/31/2026   9:33 AM              1 -- Win64 VS18  --
```

- **Configuración del puerto (8085)**

  notepad conf\httpd.conf

```
#
# Listen: Allows you to bind Apache to specific IP addresses and/or
# ports, instead of the default. See also the <VirtualHost>
# directive.
#
# Change this to Listen on specific IP addresses as shown below to
# prevent Apache from glomming onto all bound IP addresses.
#
#Listen 12.34.56.78:80
Listen 8085
```

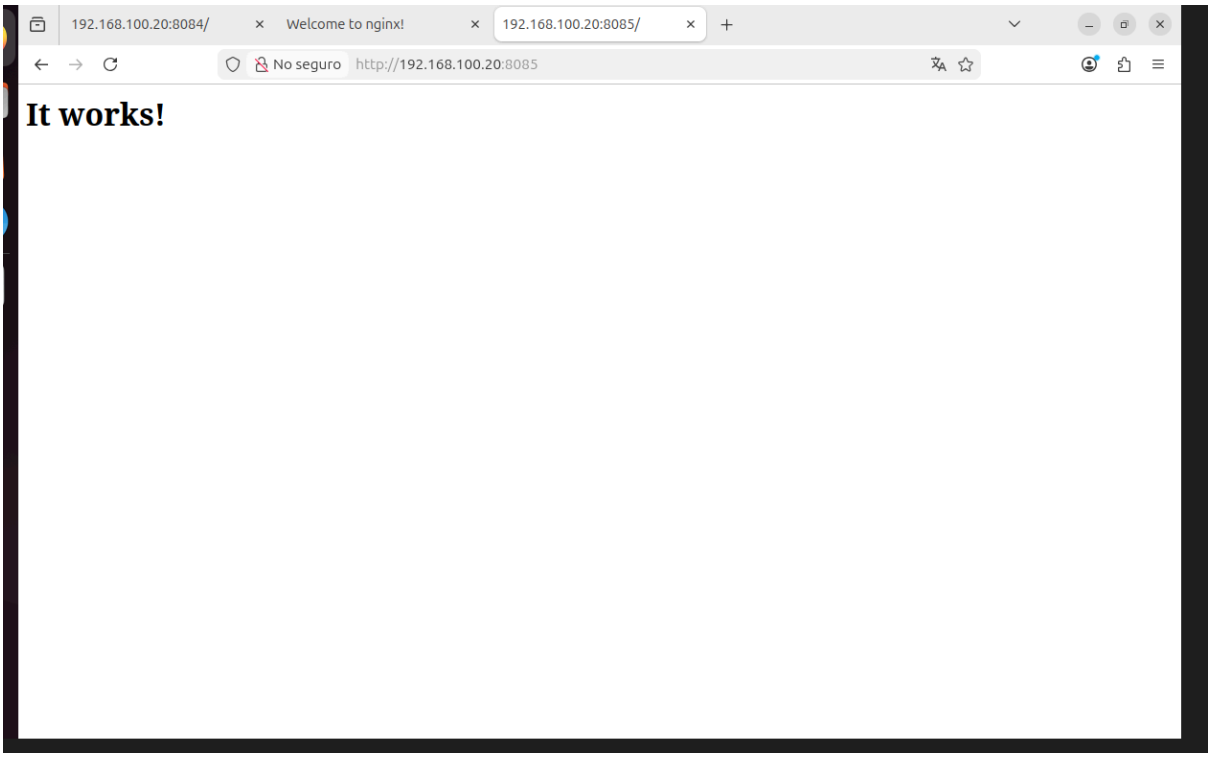
```
#
# If your host doesn't have a registered DNS name, enter its IP address here.
#
ServerName localhost:8085

#
# Deny access to the entirety of your server's filesystem. You must
# explicitly permit access to web content directories in other
# <Directory> blocks below.
#
<Directory />
    AllowOverride none
    Require all denied
</Directory>

#
# Note that from this point forward you must specifically allow
# particular features to be enabled - so if something's not working as
# you might expect, make sure that you have specifically enabled it
# below.
```

Regla

New-NetFirewallRule -DisplayName "Apache 8085" -Direction Inbound -Protocol TCP -LocalPort 8085 -Action Allow

```
PS C:\httpd-2.4.66-260223-Win64-VS18\Apache24> cd bin
PS C:\httpd-2.4.66-260223-Win64-VS18\Apache24\bin>
PS C:\httpd-2.4.66-260223-Win64-VS18\Apache24\bin>
PS C:\httpd-2.4.66-260223-Win64-VS18\Apache24\bin> .\httpd.exe
PS C:\httpd-2.4.66-260223-Win64-VS18\Apache24\bin> New-NetFirewallRule -DisplayName "Apache 8085" -Direction Inbound -Protocol TCP -L
ocalPort 8085 -Action Allow


Name                  : {a66a27b0-fa22-42cb-8421-3c75d0198d1f}
DisplayName           : Apache 8085
Description           :
DisplayGroup          :
Group                 :
Enabled               : True
Profile               : Any
Platform              : {}
Direction             : Inbound
Action                : Allow
EdgeTraversalPolicy   : Block
LooseSourceMapping    : False
LocalOnlyMapping      : False
Owner                 :
PrimaryStatus         : OK
Status                : The rule was parsed successfully from the store. (65536)
EnforcementStatus     : NotApplicable
PolicyStoreSource     : PersistentStore
PolicyStoreSourceType : Local
```

- **Prueba desde cliente**



Nginx

- **Instalación**

```
PS C:\Users\Administrator> cd \\VBOXSVR\Practica_HTTP
PS Microsoft.PowerShell.Core\FileSystem::\\VBOXSVR\Practica_HTTP> ls


    Directory: \\VBOXSVR\Practica_HTTP


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        4/24/2026   3:34 PM                documentacion
d-----        4/24/2026   5:03 PM                evidencias
d-----        4/24/2026   3:33 PM                scripts
-a----        4/24/2026   8:57 PM        2768936 nginx-1.30.0.zip
-a----        4/24/2026   3:40 PM             36 README.md


PS Microsoft.PowerShell.Core\FileSystem::\\VBOXSVR\Practica_HTTP> mkdir C:\nginx


    Directory: C:\


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        4/24/2026   9:02 PM                nginx


PS Microsoft.PowerShell.Core\FileSystem::\\VBOXSVR\Practica_HTTP> copy nginx-1.30.0.zip C:\nginx
PS Microsoft.PowerShell.Core\FileSystem::\\VBOXSVR\Practica_HTTP> cd C:\nginx
PS C:\nginx> Expand-Archive nginx-1.30.0.zip
PS C:\nginx> cd nginx-1.30.0
```

- **Configuración del puerto (8086)**

nginx.conf - Notepad

File Edit Format View Help

```
server {
    listen       8086;
    server_name  localhost;

    #access_log  logs/host.access.log  main;

    location / {
        root   html;
        index  index.html index.htm;
    }
```

Regla

New-NetFirewallRule -DisplayName "NGINX 8086" -Direction Inbound -Protocol TCP -LocalPort 8086 -Action Allow

```
PS C:\Users\Administrator> New-NetFirewallRule -DisplayName "NGNIX8086" -Direction Inbound -Protocol TCP -LocalPort 8086 -Action Allo
w


Name                  : {45f26904-c675-4f9d-aed5-1c34f52d03e5}
DisplayName           : NGNIX8086
Description           :
DisplayGroup          :
Group                 :
Enabled               : True
Profile               : Any
Platform              : {}
Direction             : Inbound
Action                : Allow
EdgeTraversalPolicy   : Block
LooseSourceMapping    : False
LocalOnlyMapping      : False
Owner                 :
PrimaryStatus         : OK
Status                : The rule was parsed successfully from the store. (65536)
EnforcementStatus     : NotApplicable
PolicyStoreSource     : PersistentStore
PolicyStoreSourceType : Local
```

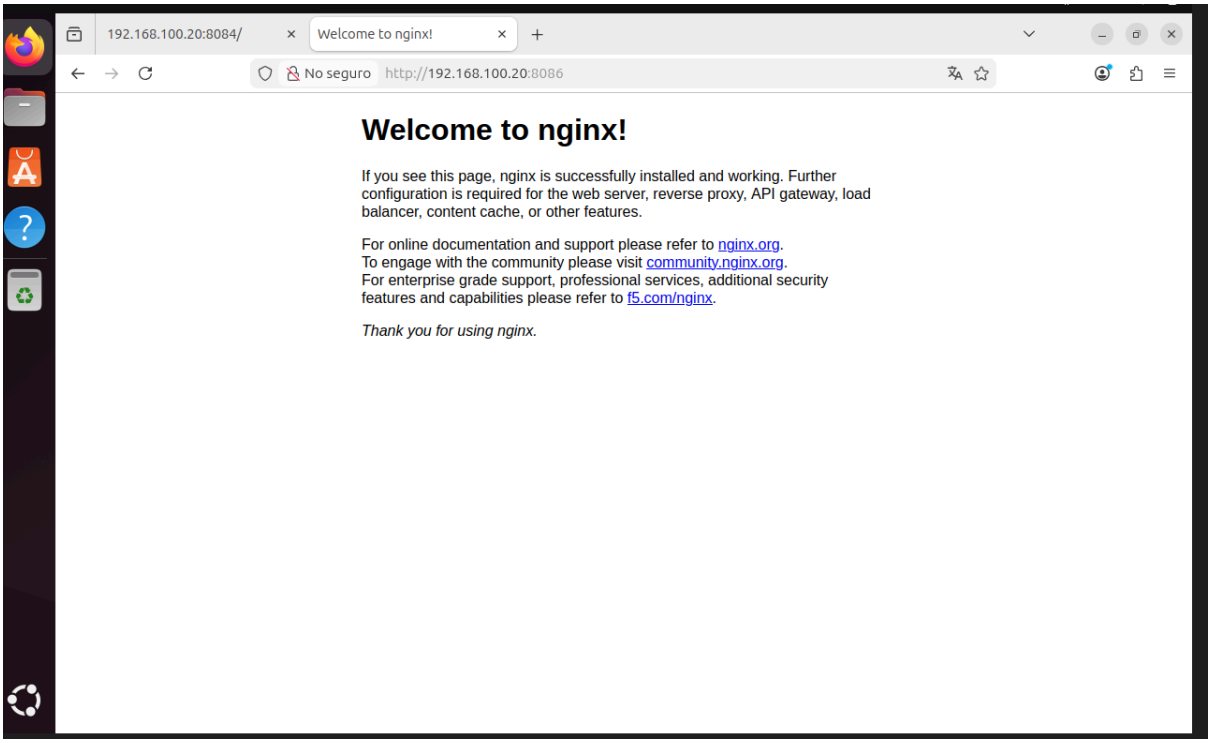
netstat -ano | findstr :8086

```
PS C:\Users\Administrator>
PS C:\Users\Administrator> netstat -ano | findstr :8086
  TCP    0.0.0.0:8086           0.0.0.0:0              LISTENING       5080
  TCP    0.0.0.0:8086           0.0.0.0:0              LISTENING       4716
PS C:\Users\Administrator>
```

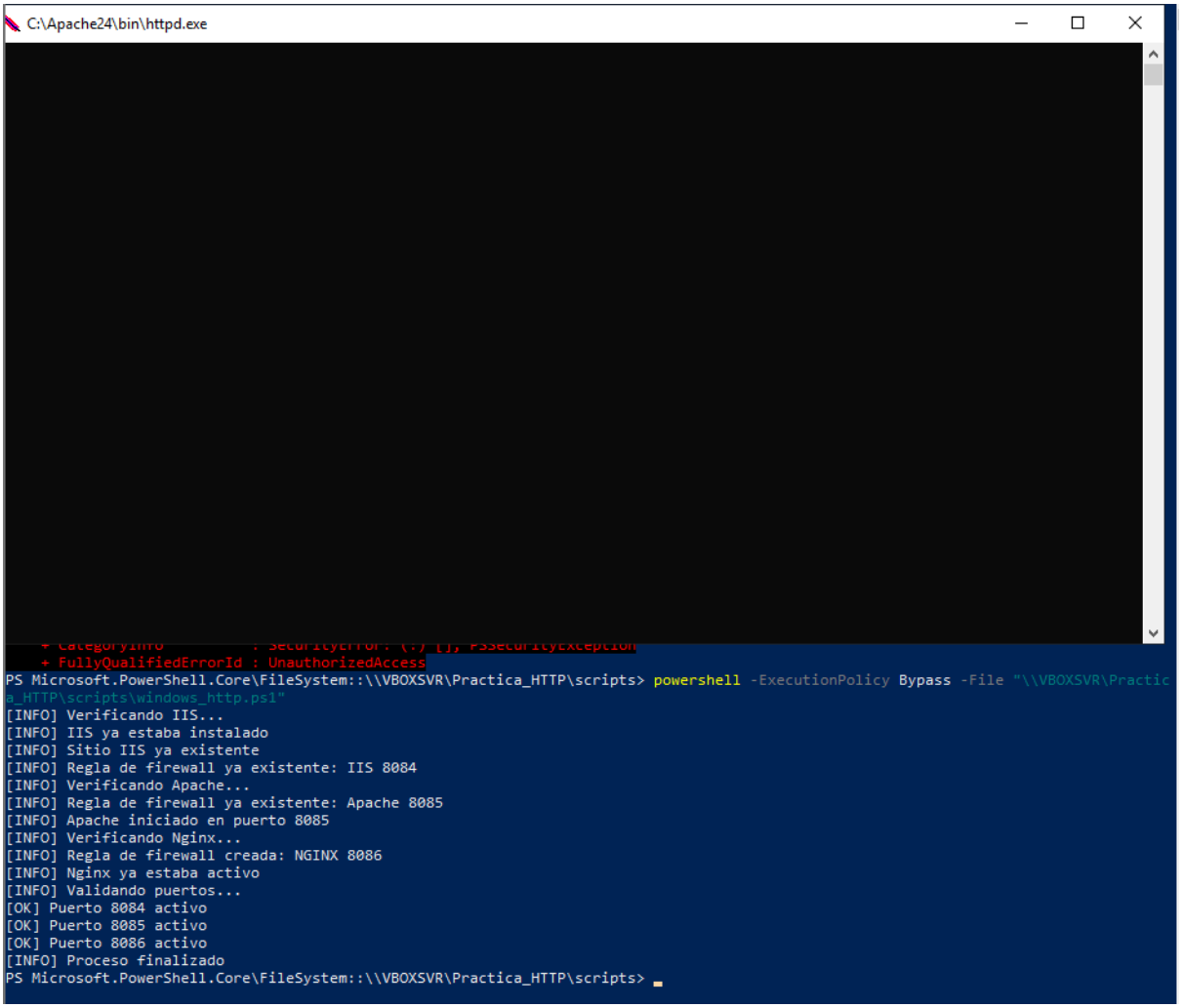
- **Prueba**

  Prueba desde cliente

http://192.168.100.20:8086



Ejecución script



Repositorio Git

https://github.com/zairazai/Practica_HTTP.git

Referencias

Apache Software Foundation. (2024). Apache HTTP Server documentation. https://httpd.apache.org/docs/

NGINX, Inc.. (2024). NGINX documentation. https://nginx.org/en/docs/

Microsoft. (2024). Internet Information Services (IIS) documentation. https://learn.microsoft.com/en-us/iis/